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Linear D. Eqⁿ

$$\frac{dy}{dx} + \underline{P(x)}y = \underline{Q(x)} \quad \checkmark$$

Solⁿ

$$y e^{\int P dx} = \int e^{\int P dx} Q dx + C$$

→ X

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$$\frac{dx}{dy} + P(y)x = Q(y)$$

$$x e^{\int P dy} = \int e^{\int P dy} Q dy + C$$

Solve $\frac{dy}{dx} + 2y \tan x = \sin x$

$$\frac{dy}{dx} + p(x)y = q(x)$$

$$p = 2 \tan x$$

$$q = \sin x$$

Solⁿ $y e^{\int p dx} = \int e^{\int p dx} q dx + C$ (1)

$$e^{\int p dx} = e^{\int 2 \tan x dx} = e^{2 \log \sec x} = \sec^2 x$$

from (1)

$$y \sec^2 x = \int \sec^2 x \sin x dx + C$$

$$= \int \sec x \frac{\sin x}{\cos x} dx + C$$

$$= \int \sec x \tan x dx + C$$

$$= \sec x + C$$

Solve $(1+y^2)dx = (\tan^{-1}y - x)dy$

$$\frac{dx}{dy} + P(y)x = Q(y)$$

$$\frac{dx}{dy} = \frac{\tan^{-1}y}{1+y^2} - \frac{x}{1+y^2}$$

$$\frac{dx}{dy} + \frac{x}{1+y^2} = \frac{\tan^{-1}y}{1+y^2}$$

$$P = \frac{1}{1+y^2}$$

$$Q = \frac{\tan^{-1}y}{1+y^2}$$

$$x e^{\int P dy} = \int e^{\int P dy} Q dy + C$$

$$e^{\int P dy} = e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1}y}$$

$$x e^{\tan^{-1}y} = \int e^{\tan^{-1}y} \left(\frac{\tan^{-1}y}{1+y^2} \right) dy + C$$

$$\tan^{-1}y = t \Rightarrow \frac{1}{1+y^2} dy = dt$$

$$= \int e^t + dt + C$$

$$= \int t e^t dt + C$$

$$x e^{\tan^{-1}y}$$

$$= (t) e^t - (1) e^t + C$$

$$= \tan^{-1}y e^{\tan^{-1}y} - e^{\tan^{-1}y} + C$$

$$\textcircled{y} e^{\int p dx} = \int e^{\int p dx} q dx + c$$

$$\frac{dy}{dx} + p(x)y = q(x)$$

$$\text{Solve } dr + (2r \cot \theta + \sin 2\theta) d\theta = 0$$

$$\frac{dr}{d\theta} + p(\theta)r = q(\theta)$$

$$\frac{dr}{d\theta} + 2r \cot \theta + \sin 2\theta = 0$$

$$\frac{dr}{d\theta} + (2r \cot \theta) r = -\sin 2\theta$$

$$P = 2 \cot \theta$$

$$Q = -\sin 2\theta$$

$$\text{Soln. } r e^{\int P d\theta} = \int e^{\int P d\theta} Q d\theta + C \quad \text{--- (1)}$$

$$e^{\int P d\theta} = e^{\int 2 \cot \theta d\theta} = e^{2 \log \sin \theta} = e^{\log \sin^2 \theta} = \sin^2 \theta$$

from (1)

$$r \sin^2 \theta = \int \sin^2 \theta (-\sin 2\theta) d\theta + C$$

$$= -\int 2 \sin^3 \theta \cos \theta d\theta + C$$

$$\sin \theta = t$$

$$\cos \theta d\theta = dt$$

$$= -2 \int t^3 dt + C = -2 \frac{t^4}{4} + C$$

$$r \sin^2 \theta = -\frac{\sin^4 \theta}{2} + C$$

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$\left(f'(y) \frac{dy}{dx} \right) + f(y)P(x) = Q(x)$$

$$f(y) = v$$

$$f'(y) \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{dv}{dx} + P(x)v = Q(x)$$

soln.

$$v e^{\int P dx} = \int e^{\int P dx} Q dx + C$$

$$\frac{f'(x) dx}{dy} + P(y) f(x) = Q(y) \quad \xrightarrow{V=f(y)}$$

another form

Equation reducible to Linear DE

Suppose eqⁿ is of the form.

$$f'(y) \frac{dy}{dx} + \underline{p(x)} f(y) = \underline{Q(x)}$$

$$\frac{dy}{du} + p(u)y = Q(u)$$

put, $f(y) = v$

diff-wrt x

$$f'(y) \frac{dy}{dx} = \frac{dv}{dx}$$

$$v \frac{dv}{dx} + p(x)v = Q(x)$$
$$v e^{\int p dx} = \int e^{\int p dx} Q dx + c$$

where $v = f(y)$ ✓

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$$f'(x) \frac{dx}{dy} + p(y) f(x) = Q(y)$$

$$f(x) = v \Rightarrow f'(x) \frac{dx}{dy} = \frac{dv}{dy} \Rightarrow \frac{dv}{dy} + p(y)v = Q(y)$$

$$v e^{\int p dy} = \int e^{\int p dy} Q dy + c$$

$$f'(y) \frac{dy}{dx} + p(x)f(y) = Q(x)$$

Solve $3y^2 \frac{dy}{dx} + 2y^3 x = 4x^3 e^{x^2}$

$$y^3 = v$$

$$3y^2 \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{dv}{dx} + 2xv = 4x^3 e^{x^2}$$

$$P = 2x$$

$$Q = 4x^3 e^{x^2}$$

solⁿ is $v e^{\int P dx} = \int e^{\int P dx} Q dx + C$ — (1)

$$e^{\int P dx} = e^{\int 2x dx} = e^{x^2}$$

from (1) $v e^{x^2} = \int e^{x^2} 4x^3 e^{x^2} dx + C$

$$= 4 \int e^{2x^2} \underbrace{(x^3)} dx + C$$

$$x^2 = t$$

$$2x dx = dt$$

$$= 4 \int e^{2t} t \frac{dt}{2}$$

$$= 2 \int t e^{2t} dt$$

$$= 2 \left[(t) \left(\frac{e^{2t}}{2} \right) - 1 \left(\frac{e^{2t}}{2 \times 2} \right) \right]$$

$$v e^{x^2}$$

$$y^3 e^{x^2}$$

$$= 2 \left[x^2 \frac{e^{2x^2}}{2} - \frac{e^{2x^2}}{4} \right]$$

$$x^2 \cdot \underline{\underline{x dx}}$$

$$f'(x) \frac{dx}{dy} + P(y) f(x) = Q(y)$$

Solve $y \frac{dx}{dy} = x + yx^2 \log y$

$$\frac{dx}{dy} = \frac{x}{y} + x^2 \log y$$

$$\frac{1}{x^2} \frac{dx}{dy} = \frac{1}{xy} + \log y$$

$$\frac{1}{x^2} \frac{dx}{dy} - \frac{1}{xy} = \log y$$

$$-\frac{1}{x} = v$$

$$\frac{1}{x^2} \frac{dx}{dy} = \frac{dv}{dy}$$

$$\frac{dv}{dy} + \frac{1}{y} v = \log y$$

$$P = \frac{1}{y}$$

$$Q = \log y$$

So $v e^{\int P dy} = \int e^{\int P dy} Q dy + C$

$$-\frac{1}{x} e^{\log y} = \int e^{\log y} \log y dy + C$$

Ans

$$y \sin x \frac{dx}{dy} - \cos x = 2y^3 \cos^2 x$$

$$f(x) \frac{du}{dy} + p(y) f(x) = q(y)$$

$$\sin x \frac{dx}{dy} - \frac{\cos x}{y} = 2y^2 \cos^2 x$$

$$\frac{\sin x}{\cos^2 x} \frac{dx}{dy} - \frac{1}{y \cos x} = 2y^2$$

$$\tan x \sec x \frac{dx}{dy} - \frac{1}{y} \sec x = 2y^2$$

$$\text{put } \sec x = v$$

$$\sec x \tan x \frac{dx}{dy} = \frac{dv}{dy}$$

$$\frac{dv}{dy} - \frac{1}{y} v = 2y^2$$

$$v e^{\int p dy} = \int e^{\int p dy} q dy + C$$

$\sec x$
 $P = -\frac{1}{y}$
 $Q = 2y^2$
Ans

Solve: $\frac{dy}{dx} - 1 = xe^{-y}$

$f(y) \frac{dy}{dx} + P(x) f(y) = Q(x)$

$$\frac{dy}{dx} - 1 = \frac{x}{e^y}$$

$$e^y \frac{dy}{dx} - e^y = x$$

$$e^y = v \Rightarrow e^y \frac{dy}{dx} = \frac{dv}{dx}$$

$$\frac{dv}{dx} - v = x$$

$$P = -1$$

$$Q = x$$

Soln

$$v e^{\int P dx} = \int e^{\int P dx} Q dx + C$$

$\downarrow$
 v
 $\downarrow$
 e^y

$$\frac{dy}{dx} + \frac{1}{x} \tan y = \frac{1}{x^2} \tan y \sin y$$

$$\frac{1}{\tan y \sin y} \frac{dy}{dx} + \frac{1}{x \sin y} = \frac{1}{x^2}$$

$$\cot y \operatorname{cosec} y \frac{dy}{dx} + \frac{1}{x} \operatorname{cosec} y = \frac{1}{x^2}$$

$$\operatorname{cosec} y = v$$

$$- \operatorname{cosec} y \cot y \frac{dy}{dx} = \frac{dv}{dx}$$

$$- \frac{dv}{dx} + \frac{1}{x} v = \frac{1}{x^2}$$

$$\frac{dv}{dx} - \frac{1}{x} v = -\frac{1}{x^2}$$

$$P = -\frac{1}{x} \quad Q = -\frac{1}{x^2}$$

HW: Solve $e^x(x+1)dx + (y^2e^{2y} - xe^x)dy = 0$

Solve $\frac{dy}{dx} = e^{x-y}(e^x - e^y)$

$$\frac{dy}{dx} = \frac{e^x}{e^y} (e^x - e^y)$$

$$\frac{dy}{dx} = \frac{e^{2x}}{e^y} - e^x$$

$$e^y \frac{dy}{dx} = e^{2x} - e^x e^y$$

$$e^y \frac{dy}{dx} + e^x e^y = e^{2x}$$

$$\frac{dy}{dx} = \frac{y^3}{e^{2x} + y^2}$$